ABSTRACT

Connected and Autonomous Vehicles (CAVs) and Reinforcement Learning (RL)-based Adaptive Traffic Signal Control (ATSC) systems have emerged as promising technologies for improving urban traffic operations. This study investigates the impact of different CAV penetration levels on traffic efficiency and traffic safety and evaluates the performance of an RL-based ATSC system under varying traffic conditions and demand patterns. The analysis was conducted using the microscopic traffic simulation software SUMO in two network scenarios: a synthetic Grid 4x4 network and a real-world Dublin City Centre (DCC) network. Three traffic conditions were considered: the HDV condition (100% HDVs), mix condition (50% HDVs and 50% CAVs), and CAV condition (100% CAVs).

Traffic efficiency was assessed using the Travel Time Rate (TTR), while traffic safety was evaluated through conflict analysis based on Post-Encroachment Time (PET) surrogate safety measures. The results show a consistent improvement in both traffic efficiency and safety with increasing CAV penetration. Across all evaluated cases, the HDV condition exhibited the highest congestion levels and conflict counts, whereas the CAV condition achieved the lowest TTR values and the greatest reduction in traffic conflicts. In the Grid 4x4 network, the CAV condition produced near-free-flow traffic conditions across most links, while in the DCC network substantial efficiency improvements were observed, particularly under free-flow conditions. In terms of safety, the CAV condition reduced traffic conflicts by up to 86.5% in the Grid 4x4 network (from 37 to 5 conflicts) and by up to 79.2% in the DCC network (from 538 to 112 conflicts) compared to the HDV condition. However, under high-demand scenarios, localized congestion and conflicts persisted even with full CAV penetration. To evaluate ATSC performance, an RL-based IDQN model was trained and tested in the Grid 4x4 network under six different traffic demand cases. The model was evaluated using average travel time and recovery time and compared with a conventional non-RL FixedTime controller. The IDQN model consistently outperformed the FixedTime controller across all traffic conditions, achieving travel time reductions of up to 30.6% in the HDV condition and 34.3% in the CAV condition. Recovery times ranged from 1 to 8 hours, demonstrating successful adaptation to varying traffic patterns and demand conditions.

Overall, the findings indicate that increasing CAV penetration can significantly enhance traffic efficiency and safety, while the integration of CAV technology with RL-based traffic signal control can further improve network performance. These results highlight the potential of intelligent transportation systems to support safer and more efficient urban mobility.